

The Role of Governance Quality in the Debt Sustainability of Chinese vs. Multilateral Loans

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List of abbreviations

ADB	Asian Development Bank
AfDB	African Development Bank
BRI	Belt and Road Initiative
CRS	Creditor Reporting System (OCED)
DSR	Debt Service Ratio
FE	Fixed Effects
GDP	Gross Domestic Product
GNI	Gross National Income
HHI	Herfindahl-Hirschman Index
IBRD	International Bank for Reconstruction and Development
IDA	International Development Association
IDB	Inter-American Development Bank
IGO	International Governmental Association
INGO	International Non-Governmental Association
IV	Instrumental Variable
PCA	Principal Component Analysis
RE	Random Effects
RMSE	Root Mean Squared Error
SD	Standard Deviation
USD	United States Dollar
WGI	Worldwide Governance Indicators

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Abstract

Qualitative analyses and case studies have dominated debates about Chinese versus multilateral lending and debt sustainability. This paper offers a systematic quantitative comparison of how governance quality affects debt service capacity across both lender types in 80 countries (2000-2023), using a harmonized dataset constructed from AidData and OECD CRS flows. Using a fixed-effects panel regression supported by extensive robustness checks, results indicate that, while governance quality is associated with the sustainability of foreign debt, Chinese and multilateral lenders demonstrate statistically indistinguishable relationships with debt sustainability across all specifications, regardless of local governance. In addition, stronger governance quality is found to be associated with higher debt service ratios, suggesting a more complex relationship between institutional quality and debt management than previously understood. These findings lead to the conclusion that, despite differences in loan allocation, debt sustainability depends more on economic and institutional fundamentals than on lender-specific characteristics.

1 Introduction

With the rise of international institutions in the aftermath of the Second World War, a similar increase in the prevalence of international development loans has taken place. While officially driven by humanitarian goals, the allegation that countries offer such loans out of selfish, geopolitical reasons has existed as long as these financing structures themselves. In recent years, a powerful narrative of “debt-trap diplomacy” has taken the spotlight, particularly among Western policy scholars raising such allegations against China’s Belt and Road Initiative (BRI). These accusations fundamentally claim that China intentionally uses development finance to create unsustainable debt burdens in strategic countries it lends to. However, this discussion has largely been built on qualitative case studies, and not on systematic data analyses. It remains an open question whether these alleged patterns exist as a systematic and quantitative relationship between a specific loan source and weaker debt outcomes.

A key theoretical mechanism in this debate is borrower governance quality, with critics suggesting that Chinese loans may be particularly detrimental to poorly governed countries, while multilateral loans use conditionality frameworks to mitigate this risk. Such conditionality frameworks are generally requirements that borrowing countries must meet to be eligible for financing. This paper moves beyond individual cases to provide a systematic and global test of the debt-trap diplomacy hypothesis, and asks the simpler, more fundamental question of

whether there is any observable statistical evidence that Chinese loans exhibit a systematically different relationship with debt sustainability compared to loans from multilateral organizations after controlling for governance quality and macroeconomic fundamentals. In this, the hypothesis that differing loan sources are associated with stronger or weaker debt outcomes is empirically tested.

By constructing a robust, filtered panel dataset and employing a multivariate regression framework designed to isolate these relationships, this study introduces a critical empirical benchmark for future research and comparisons, with even null results providing a valuable insight into the existence of alleged trends. While these findings represent statistical correlations rather than robust causality, they provide the crucial first step of determining whether any alleged differences in debt sustainability outcomes are supported by empirical analysis.

2 Literature Review

2.1 Measuring Debt Sustainability

Traditional analysis of debt levels at the country level often relies on comparing debt-to-GDP ratios, as this metric concisely shows the level of debt a government has outstanding, adjusted for economy size. However, this measurement is not optimal for the purpose of measuring how sustainable this debt is, as this ratio does not attempt to capture repayment ability, with metrics such as the debt service ratio (DSR) achieving this more effectively. Indeed, the DSR (measured as interest payments divided by exports) can serve as a useful “early warning” signal for impending banking crises (Drehmann & Juselius, 2012). Debt-to-GDP ratios do not accomplish this, with Kronick & Ambler (2023) arguing that the DSR more accurately portrays a country’s true level of financial distress. Moreover, the DSR is an established measurement for determining repayment capacity, with its deterioration increasing the probability of default (Mugun, 2021). In addition to stronger data coverage compared to alternative measurements, these reasons drive this paper’s reliance on the DSR as the primary measure for debt sustainability.

2.2 Two Philosophies of Development Finance

2.2.1 The Multilateral Model

Rather than using purely bilateral financing models, international organizations have developed institutions which facilitate multilateral financing. Rodrik (1995) presents two motivations for

this: both as a form of information provision for the private market, as well as the ability to offer financing in exchange for institutional change under loan conditionality. This channel of information provisions functions by simply using information advantages that international organizations may have, and then correcting this information asymmetry by offering finance at a price which acts as a signal to the private market (Hagen, 2009). Additionally, they can act as a stabilizing mechanism in emerging markets, which may have incomplete capital markets (Bru Muñoz, 2023). The conditionality mechanism is far more controversial, with some studies concluding that its effectiveness shows diminishing returns (Smets & Knack, 2016). A potential reason for this is the phenomenon of “isomorphic mimicry”, in which states seek to improve their surface-level governance metrics (or equivalent) without undertaking the necessary structural reform for a sustainable, long-term improvement (Andrews et al., 2017). This leaves the multilateral finance model as one built on rules-based mechanisms, whose disputed effectiveness will be further discussed in section 2.3.

2.2.2 The Chinese Model

Chinese development finance is a subject of immense political controversy, with many competing theories. The allegation of geopolitical motivations within its loan allocation is not entirely unfounded, with a “loans-for-resources” model in some developing countries (Alves, 2013), and often based on the borrower’s recognition of Taiwan (Yoo, 2021). However, it is not a purely political machine, with Yoo still concluding that “Chinese aid allocation does not only reflect its political and economic interests,” particularly prevalent through IGOs and INGOs. This introduces Chinese development finance as a contrasting model compared to that of multilateral institutions. Instead of a normative, rules-based approach, Chinese finance is explicitly “non-interventionist” in domestic affairs, providing borrowing countries with an incentive not to comply with multilateral conditionality frameworks by offering a credible alternative (Watkins, 2022). Thus, the literature discussed so far paints these two systems as competing philosophies, which at the surface level offers a clear, testable hypothesis by comparing the debt outcomes of each.

2.3 Nuances in the Conditionality Debate

A key motivation for conditionality in the multilateral framework is that of a “responsible borrower” who is more likely to use loans to effectively develop their economy if there are strings attached. Indeed, worse governance has been found to lead to higher debt levels, particularly

in low-income countries (Imaginário & Guedes, 2020). However, the effectiveness of this conditionality in actually fostering better governance is contested. Santiso (2004) argues that this is not the case, and that, even if these conditions are good policies in theory, they must be reformed to effectively achieve their goals. An additional hurdle faced by measuring the results brought by conditionality is the inherent limitations of measuring governance quality. The academic standard metric for this is the World Bank's Worldwide Governance Indicators (WGI), which this paper uses and discusses further in sections 3.1.3 and 4.2.2, but the WGI results often involve wide confidence intervals and moderate margins of error (Kaufmann et al., 2011). It is therefore important to recognize that, as in many aspects of development economics, there is no solid and indisputable data point from which governance quality, and thus the effectiveness of conditionality measured with the intent of improving it, can be determined. The effectiveness of conditionality is left as a question which this paper does not seek to definitively answer beyond offering an additional data point to inform future discussion.

At the same time, it is easy to imagine a comparison between multilateral and Chinese loans as one between using conditionality and not doing so. This would not be a fair or accurate comparison. While Chinese dialogue surrounding development finance emphasizes local sovereignty rather than imposing external values, portraying this as an abandonment of conditionality is only valid in specific areas. Rather, their conditionality mechanisms tend to operate more indirectly, as practical consequences of increased bilateral trade, which can be seen as a path-dependence mechanism (Mattlin & Nojonen, 2015). A comparison between Chinese and multilateral lending is therefore not comparing the existence of conditionality with its absence, but rather two different forms of conditionality. The two systems are also not completely disjointed from each other, with many countries receiving both multilateral and Chinese development funds, and must therefore follow two independent conditionality systems. While debt distress as a result of failed Chinese-projects does not typically cause borrowers to turn to the IMF for bailouts, severe external shocks in addition to non-performing projects can change this outcome (Kern & Reinsberg, 2022). Both lending systems and philosophies are thus intertwined enough to complicate causal analysis.

2.4 The Empirical Path Forward

There is no shortage of literature engaging with the discussion of alleged Chinese debt-trap diplomacy. In this light, several authors come to the conclusion that allegations of debt-trap diplomacy are largely a myth, by examining debt relations in specific countries (Gangte, 2020;

Himmer & Rod, 2022) and regions (Singh, 2020), finding little evidence of predatory behavior. Additionally, there is also the argument that debt-trap diplomacy is overstated as a geopolitical debate, as the tools to engage in it are not limited to governments (DeBoom, 2019). However, these either qualitative or region-limited approaches, while valuable, are inherently unable to establish a broad or systematic empirical pattern. This paper provides the crucial next step, with a global panel dataset and a multivariate regression framework that controls local governance and macroeconomic fundamentals, while also examining whether governance is associated with any lender-specific effects.

3 Data and Methodology

3.1 Data Sources and Sample Construction

3.1.1 Primary Data Sources

Data for this analysis were sourced exclusively from existing datasets, with loan flows data being systematically filtered and processed to improve usability. Multilateral loan flows were sourced from the OECD's Creditor Reporting System (CRS) (Organisation for Economic Co-operation and Development (OECD), 2025), which contains a detailed list of nearly all international credit flows. Chinese flows were sourced from AidData's Global Chinese Development Finance Dataset (Custer et al., 2023; Dreher et al., 2022), a project which constructs an approximate database on Chinese loan flows at the project level. Further variables, such as governance indicators, debt service ratio, and macroeconomic controls, were sourced from World Bank datasets.

3.1.2 Data Cleaning and Sample Selection

As the AidData and CRS databases do not immediately present a like-for-like comparison, I applied consistent filtering criteria to both datasets to create comparable flow data.

First, I restricted the analysis to debt-creating infrastructure loans by selecting relevant sector codes (140: Transport and Storage, 210-250: Economic Infrastructure, 310-330: Production Sectors). This excluded social sectors, humanitarian aid, and debt relief, as including these would have damaged consistency by including conceptually distinct observations, such as COVID-19 aid or debt-servicing assistance, the latter of which would likely have biased correlation estimates significantly.

For multilateral flows, I filtered data to only include loans from major development banks including the World Bank (e.g. IDA/IBRD), regional development banks (e.g. ADB, AfDB, IDB), and climate funds. Chinese flows were filtered to development-oriented projects using AidData's "Intent" classification.

Both sets offered deflated loan sums in constant USD, however the AidData values used 2021 as a reference year, while CRS used 2023. I therefore adjusted all AidData values to 2023 USD using U.S. Consumer Price Index conversion factors. The final loan flow variables represent real 2023 USD, with values in millions. The final, cleaned datasets account for approximately \$141 billion in Chinese development loans and \$440 billion in multilateral loans, across 80 countries between 2000-2023, giving a total of $N=1540$ country-year observations.

3.1.3 Variable Construction

The dependent variable used to represent debt sustainability is the debt service ratio (DSR) (World Bank, 2025d), measured as a country's interest payments as a percent of its exports. Several aspects support the use of this metric, most notably the notion that the DSR more accurately predicts impending financial crises by being more closely associated with the financial situations that cause them compared to the debt/GDP ratio (Kronick & Ambler, 2023). Additionally, data coverage was significantly stronger than other debt sustainability metrics, particularly in comparison to market-based measures such as the spreads of government bonds and credit default swaps.

The primary independent variables measured were the respective loan disbursements of China and multilateral sources a given country received in a year, a composite WGI index as a measure of governance quality (World Bank, 2025g), as well as interaction terms between both loan variables and governance quality. The inclusion of interaction terms is motivated by the combination of two established concepts. First, institutional quality is a known determinant of a country's debt management (Imaginário & Guedes, 2020), and secondly that Chinese and multilateral lenders exhibit notable differences in their approach towards engaging with governance in their loan allocation (Mattlin & Nojonen, 2015; Rodrik, 1995). Therefore, the interaction terms test the hypothesis that the effect of a loan from a specific lender is not uniform, but rather impacted by the local governance environment of the borrower.

Both loan flow variables are lagged by one year to mitigate reverse causality, while also recognizing that the debt impacts of increased loans are likely to not be felt or reported until the following year. The main regression specification uses total loan disbursements (millions

2023 USD) per country-year, although this is also scaled to the GDP of the borrowing country as a robustness check. The World Bank’s Worldwide Governance Index consists of six metrics, each ranging from -2.5 to 2.5, with each unit representing one standard deviation from the median score. As these are often strongly correlated with each other, I built a composite metric of these six using Principal Component Analysis (PCA), so that aggregate governance effects are efficiently captured by one variable (Imaginário & Guedes, 2020; Kaufmann et al., 2011; Jackson, 1991). PCA loadings and other details can be found in Table 11. The regression specification also includes two interaction terms, with the intention of detecting whether the debt impact of these loans depends on governance quality.

Control variables were chosen to control for macroeconomic conditions that affect both debt service capacity and loan allocation decisions. These metrics of the borrowing country are: its GDP growth rate (World Bank, 2025b), inflation rate (World Bank, 2025c), trade as a percentage of GDP (World Bank, 2025f), debt-to-GNI ratio (World Bank, 2025a), and its total foreign reserves (World Bank, 2025e). These cover overall economic output, local currency strength, economic openness, current debt situation, and their ease of repayment, respectively. All control variables were lagged by one year.

Additionally, I winsorized non-normalized variables to the 1st and 99th percentiles because countries (especially developing countries, which are overrepresented in this sample) may experience insolvency or other economic crises which would heavily skew trends while not representing the standard economic conditions this analysis attempts to measure. Table 3 and Table 4 depict general descriptive statistics of all variables before and after winsorization.

3.1.4 Descriptive Patterns

The final, cleaned dataset shows that both multilateral lending and Chinese lending are split into two separate “eras” during the sample period. As seen in Figure 1, multilateral lending saw a sharp increase in 2009, and remained stable at this higher level since then. The initial spike can easily be attributed to the global financial crisis of the preceding year and its higher-order effects across the world economy. The counter-cyclical mandate that many development banks carry causes them to ramp up investment in crisis years to mitigate the long-term investment effects of them, as demonstrated again following the COVID-19 pandemic (Barrowclough, 2025). However, the consistently high levels of development funding since then require further exploration. A primary driver for this is the recognition that ambitious international goals, such as the Sustainable Development Goals adopted in 2015, will require far more development financing

than had been provided in previous decades (World Bank, 2015). This is particularly relevant, given the wide existing infrastructure gap between developed and developing countries (Gurara et al., 2018).

Chinese lending, on the other hand, is broadly divided by 2013, with the introduction of the BRI. This ambitious infrastructure lending project saw larger scale lending, often in regional and strategically relevant countries, as seen in Table 5. However, it should be noted that both lending sources show no signs of high concentration among borrowers, both showing Herfindahl-Hirschman Indexes below 500, as seen in Table 6. The graph in Figure 1 shows several “spikes” in lending during the BRI era, which is due to the project-based nature of Chinese finance. As discussed in section 3.3.1, data is only available at the project level, and not at the country-year level, with disbursements assumed to take place in the starting year in this analysis. By contrast, multilateral data is more smooth due to the more detailed records of country-year financial flows.

3.2 Empirical Strategy

3.2.1 Baseline Specification

The baseline regression specification used is a two-way fixed effects model with the form:

$$\begin{aligned} DSR_{it} = & \beta_0 + \beta_1 ChinaLoans_{it} + \beta_2 MultilateralLoans_{it} + \beta_3 Governance_{it} \\ & + \beta_4 (ChinaLoans \times Governance)_{it} \\ & + \beta_5 (MultilateralLoans \times Governance)_{it} + \gamma X_{it} + \mu_i + \lambda_t + \varepsilon_{it} \end{aligned}$$

where, in country i and year t , DSR represents the debt service ratio, $ChinaLoans$ and $MultilateralLoans$ the respective loan disbursements this country received, $Governance$ represents the PCA composite WGI index, and the terms $(ChinaLoans \times Governance)_{it}$ and $(MultilateralLoans \times Governance)_{it}$ representing the interaction between each loan flow variable and governance quality. X_{it} represents a vector of control variables (discussed previously) and γ the corresponding vector of regression coefficients. μ_i and λ_t are inherent terms of a panel regression model, where μ_i represents the time-invariant characteristics of country i and λ_t the time effects the world experienced in the year t . As discussed in the following subsection, I chose to use a fixed effects (FE) specification, meaning that the terms μ_i and λ_t are themselves estimated as fixed parameters, rather than being assumed to be random on a standard normal distribution (e.g. $\mu_i \sim N(0, \sigma_\mu^2)$) in a random effects (RE) specification. The

error term, however, is assumed to follow a standard normal distribution $\epsilon_{it} \sim N(0, \sigma_\epsilon^2)$.

3.2.2 Model Specification: FE vs. RE

The use of panel data econometrics presents a technical question: the use of a fixed effects or a random effects model. This paper uses the conventional fixed effects regression model, although the complexities of studying governance quality warrant a more nuanced discussion. The primary issue stems from the complex conceptualization of governance as a time-variant or time-invariant characteristic of a given country. Since institutional changes are often gradual and with small incremental changes, these minor differences can often be interpreted as an inherent aspect of the country (time-invariant or between-country effect) rather than as the time-variant characteristic that it is. Clark & Linzer (2015) therefore note that the RE estimator yields a consistently lower root mean squared error (RMSE) when handling “sluggish” variables, such as institutional quality, as the RE model does not attempt to estimate time-invariant metrics. This concept is supported by Geller & Guedes (2017), who note that FE cannot accurately test within-country (time-variant) variation while also accounting for slow-moving characteristics, such as political institutions.

While this is a valid theoretical concern, empirical results do not support the use of an RE model in this specification, with a Hausman test soundly rejecting the null hypothesis that within-country effects are effectively random ($p < 0.01$). With a default RE specification showing nearly all variables significant to the 0.1% level, this dramatic increase in significance indicates potential bias from correlated country effects, as the Hausman test itself suggests. Despite the concern of inefficient estimation of the governance variable, the FE model still returns a statistically significant result for this metric despite inherently favoring low bias over low variance. This approach is supported by a between-effects specification, which shows that governance quality lacks a significant between-country explanation ($p = 0.135$), suggesting that there are indeed within-country effects for the FE model to detect.

Ultimately, this decision does not present a significant practical difference once HC3 standard errors are applied to both models. I use HC3 standard errors because of the potential for serially correlated errors, a common issue in macroeconomic data (Kézdi, 2004). After these are accounted for, differences between FE and RE HC3 results are largely marginal, demonstrating how, in this specification, the RE model is either overfitted or biased in its default model, or alternatively provides no substantial advantage over the FE model once HC3 standard errors are applied to both. Additionally, the statistical significance of key mean variables in a Mundlak

specification further rejects the use of the RE model.¹ The use of FE is supported by Hausman test results, the Mundlak specification, and between-effects results, making it the appropriate choice despite theoretical concerns regarding "sluggish" variables.

3.3 Limitations

3.3.1 Data Limitations

The use of AidData rather than official Chinese figures represents a methodological challenge in analyzing Chinese development finance, as China did not consistently report lending flows to international agencies during the examined period. AidData's dataset is reconstructed based on reports of actual financial commitments and disbursements, and remains the academic standard when analyzing Chinese financing flows despite its limitations.

This analysis employs the most comprehensive available data on development finance and governance, but several measurement considerations must be noted. AidData and OECD CRS use different reporting methodologies (project-level commitments vs. financial flow tracking) which I address through consistent aggregation rules. While I use project start dates to approximate the disbursement timing of Chinese loans, this approach may not fully capture the disbursement patterns of multi-year projects.

I measure governance quality using the World Bank's WGIs, which provide standardized cross-country comparability. While country-level scores represent general institutional environments rather than project-level governance, the WGI's methodology still allows for meaningful comparisons at the abstract level. This measurement strategy focuses on systemic governance quality rather than the characteristics of individual financial flows, which allows generalized comparisons.

3.3.2 Methodological Limitations

Endogeneity presents persistent challenges in debt sustainability analysis. Reverse causality introduces identification challenges, especially in combination with a non-random allocation of development loans by lenders. Given the severe data coverage issues a direct regression model already faces, I considered instrumental variable (IV) approaches outside this study's empirical scope. This is not because an IV approach would be inappropriate, but rather because of the challenges of finding a valid instrument which has strong data coverage and does not violate the

¹Robustness checks, such as the between-effects and Mundlak specifications, are further discussed in section 4.2.1

exclusion restriction of an IV approach. Instead, I address endogeneity concerns through several strategies, such as lagging independent variables, including comprehensive macroeconomic controls to limit omitted variable bias, and conducting extensive robustness checks. While these approaches strengthen causal inference, findings should be interpreted as conditional correlations rather than causal estimates.

3.3.3 Scope and Generalizability

This statistical analysis uses a sample of 80 countries over 22 years, yielding 1540 country-year observations, which provide sufficient power to detect moderate effects, but still leave the potential for type II errors in smaller relationships. The regional composition of the sample, while representative of major infrastructure loan recipients, exhibits a Sub-Saharan African focus (32.5% of countries). This regional focus appropriately captures major loan recipients, though may reduce its generalizability to other contexts. Results also exclude extreme debt-distress episodes, as winsorizing variables such as debt/GNI and inflation removes statistical outliers from the main specification. The empirical patterns identified in this paper provide a robust and quantitative foundation for future discussion. However, fully explaining the reasons behind these statistical patterns requires integrating quantitative evidence with deeper qualitative analysis of local institutional contexts, as studies mentioned in section 2.4 do (Barrett & Carter, 2010; Quibria, 2014). Overall, these results represent conditional correlations, not causal estimates.

4 Results

4.1 Main Findings

Building on the specification introduced in section 3.2, this section presents the regression outcomes and interprets their implications for debt sustainability.

4.1.1 Loan Source Effects

I find that the first two independent variables, *MultilateralLoans* and *ChinaLoans*, have near-zero effects, with low coefficients representing less than 5% of a standard deviation increase in the independent variable (2.0% and 0%), as well as no statistical significance ($p = 0.662$ and $p = 0.988$). Using this specification and dataset, there is no evidence indicating a

Table 1: Fixed Effects Regression Results (HC3 Robust Standard Errors)

Variable	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	22.1057	2.6289	8.409	<2.2e-16***
<i>ChinaLoans</i>	0.00003	0.0020	0.015	0.988
<i>MultilateralLoans</i>	0.00066	0.0015	0.437	0.662
<i>Governance</i>	1.8921	0.9263	2.043	0.0413*
<i>ChinaLoans</i> × <i>Governance</i>	-0.00034	0.00045	-0.748	0.455
<i>MultilateralLoans</i> × <i>Governance</i>	-0.00152	0.00112	-1.355	0.176
<i>GDP_Growth</i>	-0.1373	0.0693	-1.980	0.0479*
<i>Inflation</i>	-0.0150	0.0825	-0.182	0.856
<i>Trade_GDP_ratio</i>	-0.0971	0.0231	-4.210	2.72e-05***
<i>Debt_GNI_Ratio</i>	0.1155	0.0213	5.420	7.05e-08***
<i>TotalReserves</i>	-0.0380	0.0122	-3.103	0.00195**
Observations	N = 1540, n = 80, T = 2-22 (Unbalanced Panel)			
R² / Adj. R²	0.2370 / 0.1435			
Total SS / Residual SS	99,116 / 75,629			

Significance codes: *** 0.001, ** 0.01, * 0.05

Note: FE specification estimated using heteroskedasticity-consistent (HC3) standard errors. The control variables for trade/GDP, debt/GNI, and total foreign reserves remain highly significant, indicating that stronger macroeconomic fundamentals are associated with lower debt service ratios.

Table 2: Effect Sizes for One Standard Deviation Increase in Independent Variables

Variable	Coefficient	Variable SD	Effect Size	% of DSR SD
<i>Trade_GDP_ratio</i>	-0.0971	49.90	-4.843	41.2
<i>Debt_GNI_Ratio</i>	0.1155	39.72	4.587	39.0
<i>Governance</i>	1.8921	2.25	4.261	36.3
<i>TotalReserves</i>	-0.0380	66.29	-2.516	21.4
<i>GDP_Growth</i>	-0.1373	4.39	-0.603	5.1
<i>MultilateralLoans</i> × <i>Governance</i>	-0.0015	389.09	-0.592	5.0
<i>MultilateralLoans</i>	0.0007	349.91	0.230	2.0
<i>ChinaLoans</i> × <i>Governance</i>	-0.0003	485.62	-0.163	1.4
<i>Inflation</i>	-0.0150	8.18	-0.123	1.0
<i>ChinaLoans</i>	0.00003	123.56	0.004	0.0
Dependent Variable SD	11.75			

Note: Effect sizes calculated as coefficient × variable SD. The final column expresses the resulting change as a percentage of the dependent variable's standard deviation (Debt Service Ratio, DSR). Larger percentages indicate stronger substantive effects on debt sustainability.

systematic relationship between loan source and debt sustainability, with other variables dominating both in effect size and statistical significance. As further discussed in section 4.2.1, these findings persist across all robustness checks, indicating a robust null result rather than a product of statistical noise. The absence of effects makes understanding the statistical results of other variables particularly important for understanding potential associations with debt sustainability.

4.1.2 Governance Effects

The *Governance* variable presents a statistically significant relationship ($p = 0.041$) between overall governance quality and the DSR. With a fairly large coefficient representing 36.3% of a SD increase, this is also a quite large effect, which may be unintuitive, given that this variable's positive coefficient indicates that better governance is associated with a higher DSR. However, the effect size must be contextualized by the WGI's normalized scale, particularly regarding aspects as institutional as governance. A change of one SD would represent a massive institutional change that is not only unlikely to be found as a short-term change, but in the medium-term period that this sample measures as well. Due to the wide range of governance quality across countries, improving by one SD from the 50th percentile to the 84th percentile is a far more dramatic difference than a change of one SD in other variables.² Because of this, the effect size could be much smaller in practice, though still an interesting result that can be interpreted as suggestive evidence if treated with appropriate skepticism. Nonetheless, this baseline governance effect is important to account for when interpreting the subsequent interaction terms, and further implications of this coefficient are discussed in section 5.2.

4.1.3 Interaction Terms

The two interaction terms (*MultilateralLoans* × *Governance*) and (*ChinaLoans* × *Governance*), which represent a core test of this paper, show neither large coefficients (5.0% and 1.4% of a SD increase, respectively) nor statistical significance ($p = 0.176$ and $p = 0.455$, respectively). With any possible governance effects already absorbed by *Governance*, I find that there is no evidence supporting the conclusion that the effects of loan sources on debt sustainability are related to governance quality, in addition to the finding that loan sources on their own have no correlation with debt sustainability. Any strongly statistically significant findings are in-

²WGI variables are originally scaled from -2.5 to 2.5 , but this is disrupted by the PCA composite process, which widens this range. However, the values are still based on a normalized metric, which compresses large differences in governance quality into a relative scale, centered around 0.

stead found with variables which indicate overall macroeconomic health, such as trade openness and pre-existing debt levels, rather than geopolitical situations. This finding represents a strong counterargument to the claim that either lending strategy performs better in specific governance-related situations, and the implications of this will be discussed further in the next chapter.

4.1.4 Significant Control Variables

Three control variables, each representing key macroeconomic indicators, dominate effect sizes while being highly statistically significant. The trade/GDP ratio (41.2% of SD increase, $p < 0.001$) and total foreign reserves (21.4% of SD increase, $p = 0.002$) both have significant negative correlations with the debt service ratio, indicating that greater values of each are associated with stronger debt sustainability. The influence channels for these variables are also rather intuitive, as economies with greater foreign holdings can more easily pay foreign debts, as can economies more dependent on trade, with these also having better access to foreign currencies. Meanwhile, the debt-to-GNI ratio (39.0% of SD increase, $p < 0.001$) has a strong positive correlation, indicating that countries with already high debt levels will struggle to service future credits. The large effect sizes of these variables strongly support the conclusion that macroeconomic indicators are more strongly associated with a country's debt sustainability than the previously discussed loan source terms.

4.1.5 Insignificant Control Variables

Two control variables, *GDP_Growth* and *Inflation*, have neither statistically significant effects ($p = 0.200$ and $p = 0.758$, respectively) nor large effect sizes (4.8% and 1.9% of a SD increase, respectively). There are two potential explanations for this result. The first is that these variables do not have a direct channel towards debt payments in the same way that more significant control variables do. While GDP growth is a useful macroeconomic indicator, the effects of this are so broad that it may be difficult for direct associations with the debt service ratio to be detected. A large increase in exports would theoretically increase GDP growth and lower the debt service ratio,³ but this specific effect may be difficult to capture and may also be represented through other metrics, such as trade/GDP or total foreign reserves. Inflation, while relevant for foreign exchange rates, is most likely a too indirect channel to have a noticeable effect in this specification. Despite these null results, these variables are still relevant to the

³Higher exports increase GDP directly, while increasing the denominator of the debt service ratio (measured as interest payments divided by exports), decreasing the ratio's value

overall regression specification, as they ensure that other variables do not capture extraneous effects.

4.2 Robustness Framework

Multiple robustness tests confirm that the results of the main regression specification are not a result of statistical anomalies or methodological differences.

4.2.1 Model Specification Robustness

The persistence of the main results is confirmed across a variety of alternative panel data estimators. The key parameters of interest remain stable, demonstrating that the findings are robust and not driven by model selection. As seen in the Mundlak specification, shown in Table 7, core loan-source terms remain statistically insignificant in both fixed and random effects frameworks, demonstrating robustness to different modeling assumptions.⁴ The significance of the governance term strengthening slightly ($p = 0.014$), which suggests that the main FE specification may be understating this parameter's significance. The significance of the control variables remains broadly consistent with the FE model. Furthermore, the Mundlak results indicate that countries which receive more multilateral loans show different dynamics ($p = 0.01$). The significant between-country effects for multilateral loans ($p < 0.05$), shown in Table 8, supports the Mundlak result, suggesting that countries systematically selected for multilateral lending have different baseline debt characteristics. However, the consistently null within-country effects indicate that receiving multilateral loans does not, in itself, produce differential debt outcomes. This critical distinction between selection and treatment effects ultimately reinforces my main finding of null loan-source effects, while introducing an interesting avenue for future research.

4.2.2 Measurement Robustness

In addition to model-based robustness checks, I also tested my main FE specification across multiple variations of the variables themselves. A further robustness test of the governance-debt relationship was to split my country sample into “high governance” and “low governance” groups, considering all countries in the upper third of WGI scores as “high governance” and the lower third as “low governance”. This robustness check showed no significant relationship between the “high governance” dummy variable and an increased DSR, indicating that governance effects are more likely to occur on a continuous spectrum than as a discrete change (see

⁴See also: Random Effects regression results, Table 9

Table 10).

To test the robustness of the loan variables, I scaled the loan sizes to the GDP of the respective country, thereby examining the size of the received loan flows to the economy as a whole. I also adjusted the lag in the main specification by testing specifications with no lagging as well as one with a two-year lag. Results of these checks remained largely consistent with the main specification, with some fragility in governance quality's. A summary of these results, along with the temporal robustness discussed in the following paragraph, can be seen in Table 13. The results are also largely robust to controlling for regional heterogeneity, with multilateral loans showing no region-specific effects. Chinese loans do show significant region-specific effects with Europe & Central Asia and Middle East & North Africa, but the primary *ChinaLoans* variable remained insignificant in this specification, dampening interpretive relevance (see Table 14).

Temporal robustness was tested by examining any possible relationships before the launch of China's BRI in 2013. Previously null results remained while control variables remained significant, although the governance variable was no longer significant in this HC3 specification ($p = 0.313$), and the multilateral loans variable became significant at the 1% level. However, this specification suffers from a small sample size, with only $N = 803$ (52% of total sample) observations, making deeper interpretation of this result difficult.

No evidence of multicollinearity was found, with no notable results in the variable correlation matrix (See Table 11) and all variables showing a VIF of less than 3 (see Table 12). The persistence of null interaction effects across this comprehensive robustness framework provides strong evidence that loan source does not moderate governance-debt relationships, while still supporting the possibility of governance quality overall being associated with different debt relationships.

5 Discussion

5.1 Comparing Chinese and Multilateral Loans

Across several specifications, governance measures, and robustness frameworks, there is a consistent lack of evidence supporting any kind of statistical difference in the effects that Chinese or multilateral loans have on the debt sustainability of the borrower. This serves as a further point to critique the often Western-led narrative of Chinese debt-trap diplomacy, which has rightly already been viewed with academic skepticism (Himmer & Rod, 2022). A surface-level

view would be to assume that a convergence in risk management practices between Chinese and multilateral lenders has taken place, with the null result following from a genuine similarity in lending practices. Yet this conclusion is not supported by Chinese lending strategies, with their lack of the conditionality metrics used by multilateral institutions continuing across the whole time range of this sample. Additionally, an analysis of lending before and after 2013 (the launch year of the BRI) yields the same null result in both cases. Another possible explanation is that any country-specific aspects that influence loan assignment are already accounted for by control variables or between-country effects. However, it is also possible that the simplest explanation, that the lack of conditionality criteria does not negatively impact debt sustainability, also holds merit, and if so would not be a unique perspective on conditionality as a whole, as discussed in chapter 2. Despite the fundamentally different approaches to conditionality that the Chinese and multilateral lending models display, both appear to respond to risk factors similarly enough for their outcomes to be statistically indistinguishable.

While the absence of evidence is not evidence of absence alone, the persistence of this null result across specifications still represents a substantive finding in itself. The robust lack of any finding does suggest that this lack of correlation is, in fact, because these factors are not correlated with debt sustainability. This empirical pattern represents another challenge to narratives of debt-trap diplomacy, while also suggesting that Chinese and multilateral financial systems are not particularly differentiated in outcome overall.

5.2 The Governance Indicator

The regression analysis yields a high but marginally significant coefficient indicating that stronger governance scores are associated with higher debt service ratios, presenting a puzzle rather than a clear, definitive finding. After all, weaker governance has been shown to be associated with greater debt levels (Imaginário & Guedes, 2020), which could worsen debt sustainability if not handled properly. While a detailed analysis of the causes behind this result are beyond the scope of this paper, the consistent statistical result in my regression analysis invites speculative exploration of potential mechanisms to target areas of future research. This discussion does not seek to explain this correlation, but rather to find avenues for future research to demystify this relationship.

One hypothesis is that better governed countries manage their debt more responsibly, allowing them to tolerate higher servicing ratios without a risk of default, causing them to take on higher debt levels. In essence, this argues that the greater debt levels that Guedes and Imag-

inário investigated are independent from a country's ability to service this debt, drawing upon the logic of Kronick & Ambler (2023) that debt levels themselves do not predict financial distress. The implications of my statistical results could thus be further tested by investigating the association between higher debt levels and debt servicing capabilities in different governance environments.

Ultimately, this governance puzzle, while intriguing, does not undermine the primary conclusion of this paper. The relatively large but unclear effect of governance, combined with the persistent null effects of the loan source variables, solidifies the argument that domestic institutions and economic fundamentals are the primary predictors of debt sustainability. While the initial hypothesis was that loans from various sources would have different impacts based on the governance quality of the borrower, this analysis provides a robust lack of evidence to support this claim.

5.3 Between-Country vs. Within-Country Effects

Across specifications, there is a consistent null result when analyzing the impacts of different lenders. However, the between-effects result showing a selection bias should still be examined further. This specification indicates a significant correlation between countries receiving multilateral loan flows and higher debt service ratios. Similarly to the previously discussed governance indicator, this is likely due to other unmeasurable aspects associated with a higher DSR, rather than the increased debt level itself. It would be a plausible explanation to suggest that, within the LDC-skewed country sample used, a higher DSR is associated with stronger market development and transparency, making the borrower seem more credible despite its higher servicing costs. More broadly, it is simply possible that the countries which are capable of meeting multilateral conditionality requirements have different financial situations than those who cannot, supporting the notion that conditionality does not cause improved governance situations (Santiso, 2004), but rather rewards already existing ones. An additional, though speculative, explanation could involve a geopolitical angle, in which countries in greater political dialogue with multilateral institutions have different economic trajectories than those who intentionally do not, but this hypothesis is difficult to test quantitatively. While the exact causes for this selection bias warrant future research, the Mundlak specification explicitly separates selection (between-country) from treatment (within-country) effects, showing that while there is a selection effect in loan allocation, debt service ratios remain robust to influence based on loan source.

The chosen FE methodology effectively isolates within-country effects from between-country selection (see sections 3.2.2, 4.2.1), addressing concerns about unobserved heterogeneity. Despite the challenges in observing year-to-year changes in governance due to how slow these measures change over time, the FE model effectively captures meaningful year-to-year changes across the time period. The similarity in the results between FE and RE HC3 specifications demonstrates that governance is both an evolving institutional quality (rather than a fixed country effect) and a corollary with debt sustainability

5.4 Impact of Findings

5.4.1 Policy Implications

The policy implications of these findings carry distinct implications for various key actors in development finance.

Borrowers can interpret these empirical results as clear evidence that economic fundamentals are the most prominent variable determining long-term debt sustainability. Most notably, prioritizing building export capacity and foreign reserves have shown to be a more reliable path towards achieving debt sustainability than strategically selecting lenders based on their perceived conditionality systems. Additionally, strong transparency allows lenders to be aware of these fundamentals, mitigating inflated risk premiums due to information asymmetry. Transparency may also influence debt sustainability directly, and while the empirical results are not clear-cut, its economic magnitude warrants attention when discussing deeper institutional reforms. These economic and institutional factors far outweigh any lender-selection effects in aggregate.

Lenders, particularly multilateral ones, should investigate why their divergent formal risk-management methods produce convergent outcomes, potentially questioning the cost-effectiveness of complex conditionality systems. While this paper does not explicitly evaluate the effectiveness of conditionality, the marginal benefit of the ex-ante conditionality used by multilateral lenders may be less pronounced in influencing debt sustainability than previously assumed, especially when considering the notable selection effects found in multilateral lending and the similar debt outcomes to non-conditionality based systems.

Finally, the discussion of debt-trap diplomacy would greatly benefit from a deeper analysis of overall trends. Policymakers and analysts should treat narratives of systemic debt-trap diplomacy with skepticism. While it is certainly possible that individual instances may take place, this analysis finds no aggregate evidence to support the claim that any large-scale strategy exists,

or that Chinese lending creates systematically worse debt outcomes than multilateral lending.

5.4.2 Future Research Avenues

In many ways, this analysis opens more questions than it answers, particularly surrounding a borderline significant governance variable. While it is possible that the true effect is simply on the margins of having a consistent impact, its large effect size implies that there is more to be found. Despite this study's comprehensive robustness checks, it is still possible that a different methodological approach (including different variables) can more clearly explain the relationship between governance quality and debt sustainability. For example, a more microfoundational or theoretical approach could uncover the mechanisms driving the statistical relationship found in this paper, as these would not be burdened by substantial statistical noise.

The selection effects found in the Mundlak specification also warrant further investigation. Further research could seek to explain the sources of the selection effect shown by multilateral loans, for example through an instrument variable or regression discontinuity approach. Doing so could potentially find the effects of eligibility for multilateral lenders using conditionality. Quantifying selection effects more accurately allows multilateral lenders to more accurately find causal effects of their conditionality frameworks by correcting for selection bias.

Most importantly, future research can seek to explain why this analysis showed a robust null result in the loan source variables despite their noticeably different lending practices. While any study involving Chinese lending will suffer from data coverage issues, a deeper analysis that seeks to isolate causal effects from a lender's perspective (rather than a borrower's perspective as in this paper) can illuminate the mechanisms through which multilateral and Chinese organizations achieve similar statistical outcomes despite vastly different conditionality structures.

6 Conclusion

This analysis demonstrates that many aspects of the geopolitical debate surrounding Chinese loans compared to multilateral loans are not supported by observable debt outcomes. The robust statistical near-equivalence between lenders in debt outcomes points to the simpler economic reality that a country's debt servicing capacity is not determined by the lender but by the overall macroeconomic health of the borrower. Narratives of lender-specific risks find no support in this analysis. The most intriguing result of this study is rather one that it was not originally intended to resolve. The relationship between better governance and a higher debt service ratio initially seems counterintuitive, and serves as a powerful reminder that institutional quality influences

economic outcomes through complex channels that cannot be explained by simple narratives.

Despite the inherent empirical challenges of the subject, this result shifts future research priorities from questioning whether or not lenders differ in their debt outcomes to explaining why they do not. In particular, the selection bias displayed by multilateral lenders warrants further analysis, especially when considering the similar debt outcomes to Chinese loans. Deeper analysis of the causes behind this phenomenon can further educate the debate on the effectiveness of loan conditionality within the multilateral system. Ultimately, this study provides a robust empirical benchmark to establish that no differential relationship between loan sources exists, while the task of explaining why this is remains for future research.

References

- Alves, A. C. (2013, March). Chinese Economic Statecraft: A Comparative Study of China's Oil-backed Loans in Angola and Brazil. *Journal of Current Chinese Affairs*, 42(1), 99–130. Retrieved from <https://journals.sagepub.com/doi/10.1177/186810261304200105> doi: 10.1177/186810261304200105
- Andrews, M., Pritchett, L., & Woolcock, M. (2017, January). Looking like a state: The seduction of isomorphic mimicry. In M. Andrews, L. Pritchett, & M. Woolcock (Eds.), *Building State Capability: Evidence, Analysis, Action* (p. 0). Oxford University Press. Retrieved from <https://doi.org/10.1093/acprof:oso/9780198747482.003.0003> doi: 10.1093/acprof:oso/9780198747482.003.0003
- Barrett, C. B., & Carter, M. R. (2010). The Power and Pitfalls of Experiments in Development Economics: Some Non-random Reflections. *Applied Economic Perspectives and Policy*, 32(4), 515–548. Retrieved from <https://onlinelibrary.wiley.com/doi/abs/10.1093/aep/pq023> (_eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1093/aep/pq023>) doi: 10.1093/aep/pq023
- Barrowclough, D. (2025, May). Mandate Matters: Evolving Views and Counter-cyclical Surprises from the World's Newest, Southern-led, Multilateral Development Banks*. *Review of Political Economy*, 37(3), 965–989. Retrieved from <https://www.tandfonline.com/doi/full/10.1080/09538259.2025.2470192> doi: 10.1080/09538259.2025.2470192
- Bru Muñoz, M. (2023, January). *The forgotten lender: the role of multilateral lenders in sovereign debt and default* [SSRN Scholarly Paper]. Rochester, NY: Social Science Research Network. Retrieved from <https://papers.ssrn.com/abstract=4317493> doi: 10.2139/ssrn.4317493
- Clark, T. S., & Linzer, D. A. (2015, May). Should I Use Fixed or Random Effects? *Political Science Research and Methods*, 3(2), 399–408. Retrieved from <https://www.cambridge.org/core/journals/political-science-research-and-methods/article/should-i-use-fixed-or-random-effects/12DFCB222123587A37163F2226E85C67> doi: 10.1017/psrm.2014.32

- Custer, S., Dreher, A., Elston, T. B., Escobar, B., Fedorochko, R., Fuchs, A., ... Zhang, S. (2023). *Tracking chinese development finance: An application of AidData's TUFF 3.0 methodology*. Williamsburg, VA: AidData at William & Mary.
- DeBoom, M. (2019, December). Who is afraid of 'debt-trap diplomacy'? Geopolitical narratives, agency and the multiscalar distribution of risk. *Area Development and Policy*, 5, 1–8. doi: 10.1080/23792949.2019.1703556
- Dreher, A., Fuchs, A., Parks, B. C., Strange, A., & Tierney, M. J. (2022). *Banking on beijing: The aims and impacts of china's overseas development program*. Cambridge, UK: Cambridge University Press.
- Drehmann, M., & Juselius, M. (2012, September). Do Debt Service Costs Affect Macroeconomic and Financial Stability? *BIS Quarterly Review September*. Retrieved from <https://papers.ssrn.com/abstract=2206301>
- Gangte, L. (2020, April). The Debt-Trap Diplomacy Revisited: A Case Study on Sri Lanka's Hambantota Port. *Artha Journal of Social Sciences*, 19(2), 53–66. Retrieved from <https://journals.christuniversity.in/index.php/artha/article/view/2553> doi: 10.12724/ajss.53.4
- Geller, G., & Guedes, M. J. C. (2017, January). Was the collapse of the communist bloc a game changer in the stock markets? Left-wing vs. right-wing political preferences and stock market development. *Research in International Business and Finance*, 39, 423–432. Retrieved from <https://www.sciencedirect.com/science/article/pii/S0275531916301015> doi: 10.1016/j.ribaf.2016.09.002
- Gurara, D., Klyuev, V., Mwase, N., & Presbitero, A. F. (2018, February). Trends and Challenges in Infrastructure Investment in Developing Countries. *International Development Policy | Revue internationale de politique de développement*(10.1). Retrieved from <https://journals.openedition.org/poldev/2802> (ISBN: 9782940600069 Number: 10.1 Publisher: Institut de hautes études internationales et du développement) doi: 10.4000/poldev.2802
- Hagen, R. J. (2009, September). Basic analytics of multilateral lending and surveillance. *Journal of International Economics*, 79(1), 126–136. Retrieved from <https://www.sciencedirect.com/science/article/pii/S0022199609000634> doi: 10.1016/j.jinteco.2009.04.007

- Himmer, M., & Rod, Z. (2022, September). Chinese debt trap diplomacy: reality or myth? *Journal of the Indian Ocean Region*, 18(3), 250–272. Retrieved from <https://doi.org/10.1080/19480881.2023.2195280> (Publisher: Routledge _eprint: <https://doi.org/10.1080/19480881.2023.2195280>) doi: 10.1080/19480881.2023.2195280
- Imaginário, J., & Guedes, M. J. (2020). Governance and government debt. *Risk Governance and Control: Financial Markets and Institutions*, 10(3), 34–49. Retrieved from <https://www.virtusinterpress.org/Governance-and-government-debt.html> doi: 10.22495/rgcv10i3p3
- Jackson, E. (1991). PCA With More Than Two Variables. In *A Use's Guide to Principal Components* (pp. 26–62). John Wiley & Sons, Ltd. Retrieved from <https://onlinelibrary.wiley.com/doi/abs/10.1002/0471725331.ch2> (_eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1002/0471725331.ch2>) doi: 10.1002/0471725331.ch2
- Kaufmann, D., Kraay, A., & Mastruzzi, M. (2011, September). The Worldwide Governance Indicators: Methodology and Analytical Issues. *Hague Journal on the Rule of Law*, 3(2), 220–246. Retrieved from <https://www.cambridge.org/core/journals/hague-journal-on-the-rule-of-law/article/abs/worldwide-governance-indicators-methodology-and-analytical-issues1/D2DBA45F726D0442A29A651710D96C37> doi: 10.1017/S1876404511200046
- Kern, A., & Reinsberg, B. (2022, October). The Political Economy of Chinese Debt and International Monetary Fund Conditionality. *Global Studies Quarterly*, 2(4), ksac062. Retrieved from <https://doi.org/10.1093/isagsq/ksac062> doi: 10.1093/isagsq/ksac062
- Kronick, J., & Ambler, S. (2023, October). Predicting financial crises: debt versus debt service ratios. *Applied Economics*, 55(47), 5553–5571. Retrieved from <https://www.tandfonline.com/doi/full/10.1080/00036846.2022.2140112> doi: 10.1080/00036846.2022.2140112
- Kézdi, G. (2004). Robust standard error estimation in Fixed-Effects panel models. *Hungarian Statistical Review*, 82(SN9), 95–116. Retrieved from <https://real.mtak.hu/138595/> (Publisher: Central Statistical Office)

- Mattlin, M., & Nojonen, M. (2015, July). Conditionality and Path Dependence in Chinese Lending. *Journal of Contemporary China*, 24(94), 701–720. Retrieved from <https://doi.org/10.1080/10670564.2014.978154> (Publisher: Routledge _eprint: <https://doi.org/10.1080/10670564.2014.978154>) doi: 10.1080/10670564.2014.978154
- Mugun, W. (2021). Effect of Total Debt Service, Capital stock and Inflation on Economic Growth in SSA Difference GMM Analysis. *International Journal of Research and Scientific Innovation (IJRSI)*, 8 (3), 144–154.
- Organisation for Economic Co-operation and Development (OECD). (2025). *OECD creditor reporting system (CRS) database*. Retrieved 2025-01-05, from [https://data-explorer.oecd.org/vis?df\[ds\]=DisseminateFinalBoost&df\[id\]=DSD_CRS%40DF_CRS&df\[ag\]=OECD.DCD.FSD&dq=DAC..1000.100._T._T.D.Q._T..&lom=LASTNPERIODS&lo=5&to\[TIME_PERIOD\]=false](https://data-explorer.oecd.org/vis?df[ds]=DisseminateFinalBoost&df[id]=DSD_CRS%40DF_CRS&df[ag]=OECD.DCD.FSD&dq=DAC..1000.100._T._T.D.Q._T..&lom=LASTNPERIODS&lo=5&to[TIME_PERIOD]=false) (Accessed via OECD Data Explorer)
- Quibria, M. (2014, January). Aid effectiveness: research, policy and unresolved issues. *Development Studies Research*, 1(1), 75–87. Retrieved from <https://doi.org/10.1080/21665095.2014.922890> (Publisher: Routledge _eprint: <https://doi.org/10.1080/21665095.2014.922890>) doi: 10.1080/21665095.2014.922890
- Rodrik, D. (1995, June). *Why is there Multilateral Lending?* (Tech. Rep. No. w5160). Cambridge, MA: National Bureau of Economic Research. Retrieved from <http://www.nber.org/papers/w5160.pdf> doi: 10.3386/w5160
- Santiso, C. (2004). Development finance, governance and conditionality: politics matter. *International Public Management Journal*, 7(1), 73–100. (Publisher: Information Age Publishing Inc)
- Singh, A. (2020, August). The myth of ‘debt-trap diplomacy’ and realities of Chinese development finance. *Third World Quarterly*, 42(2), 239–253. Retrieved from <https://doi.org/10.1080/01436597.2020.1807318> (Publisher: Routledge _eprint: <https://doi.org/10.1080/01436597.2020.1807318>) doi: 10.1080/01436597.2020.1807318
- Smets, L., & Knack, S. (2016). World Bank lending and the quality of economic policy. *The Journal of Development Studies*, 52(1), 72–91. (Publisher: Taylor & Francis)
- Watkins, M. (2022, October). Undermining conditionality? The effect of Chinese development assistance on compliance with World Bank project agreements. *The Review of International*

- Organizations*, 17(4), 667–690. Retrieved from <https://doi.org/10.1007/s11558-021-09443-z> doi: 10.1007/s11558-021-09443-z
- World Bank. (2015). *Financing for development: World bank group support for the 2030 agenda*. Washington, DC. Retrieved from <https://thedocs.worldbank.org/en/doc/622841485963735448-0270022017/original/DC20150002EFinancingforDevelopment.pdf>
- World Bank. (2025a). *External debt stocks (% of GNI)*. Retrieved 2025-01-05, from <https://data.worldbank.org/indicator/DT.DOD.DECT.GN.ZS> (World Bank International Debt Statistics)
- World Bank. (2025b). *GDP growth (annual %)*. Retrieved 2025-01-05, from <https://data.worldbank.org/indicator/NY.GDP.MKTP.KD.ZG> (World Bank World Development Indicators)
- World Bank. (2025c). *Inflation, consumer prices (annual %)*. Retrieved 2025-01-05, from <https://data.worldbank.org/indicator/FP.CPI.TOTL.ZG> (World Bank World Development Indicators)
- World Bank. (2025d). *Total debt service (% of exports of goods, services and primary income)*. Retrieved 2025-01-05, from <https://data.worldbank.org/indicator/DT.TDS.DECT.EX.ZS> (World Bank International Debt Statistics)
- World Bank. (2025e). *Total reserves (includes gold, current us\$)*. Retrieved 2025-01-05, from <https://data.worldbank.org/indicator/FI.RES.TOTL.CD> (World Bank World Development Indicators)
- World Bank. (2025f). *Trade (% of GDP)*. Retrieved 2025-01-05, from <https://data.worldbank.org/indicator/NE.TRD.GNFS.ZS> (World Bank World Development Indicators)
- World Bank. (2025g). *Worldwide governance indicators, 2025 revision*. Retrieved from www.govindicators.org
- Yoo, E. (2021). Chinese Development Finance and its determinants: Does global governance matter? *Development Policy Review*, 39(3), 471–492. Retrieved from <https://onlinelibrary.wiley.com/doi/abs/10.1111/dpr.12514>

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Appendix

Note: Data replication files can be found online at <https://github.com/nicodrew/chinamultilaterals>

Descriptive Statistics

Table 3: Descriptive Statistics of Key Variables, winsorized

Variable	N	Mean	SD	Min	Max	Median
DSR	2,872	13.766	11.754	0.422	62.696	10.562
ChinaLoans	2,852	35.801	123.559	0.000	863.008	0.000
MultilateralLoans	2,852	146.003	349.912	0.000	2,001.573	2.970
Governance	4,415	-0.029	2.252	-5.830	4.828	-0.343
GDP Growth	5,837	3.428	4.392	-12.459	16.657	3.711
Inflation	4,460	5.873	8.182	-2.400	53.869	3.500
Trade/GDP Ratio	5,050	84.529	49.903	24.693	345.255	71.843
Debt/GNI Ratio	2,920	52.040	39.715	3.954	251.323	40.726
Total Reserves	3,275	57.085	66.289	1.414	445.666	37.358

Table 4: Descriptive Statistics of Key Variables, non-winsorized

Variable	N	Mean	SD	Min	Max	Median
DSR	2,872	13.988	13.011	0.003	134.803	10.562
ChinaLoans	2,852	49.352	280.767	0.000	6,836.191	0.000
MultilateralLoans	2,852	154.618	411.836	-1.357	5,170.949	2.970
Governance	4,415	-0.029	2.252	-5.830	4.828	-0.343
GDP Growth	5,837	3.456	5.543	-54.336	91.137	3.711
Inflation	4,460	30.588	1,055.541	-72.700	65,374.100	3.500
Trade/GDP Ratio	5,050	85.175	54.622	2.699	863.195	71.843
Debt/GNI Ratio	2,920	53.213	47.537	1.154	610.452	40.726
Total Reserves	3,275	69.615	195.983	0.009	3,840.101	37.358

Note: Some variables, most notably inflation, GDP growth, and debt/GNI, display severe outliers which risk biasing the sample

Figure 1: Aggregate lending over time, split by source

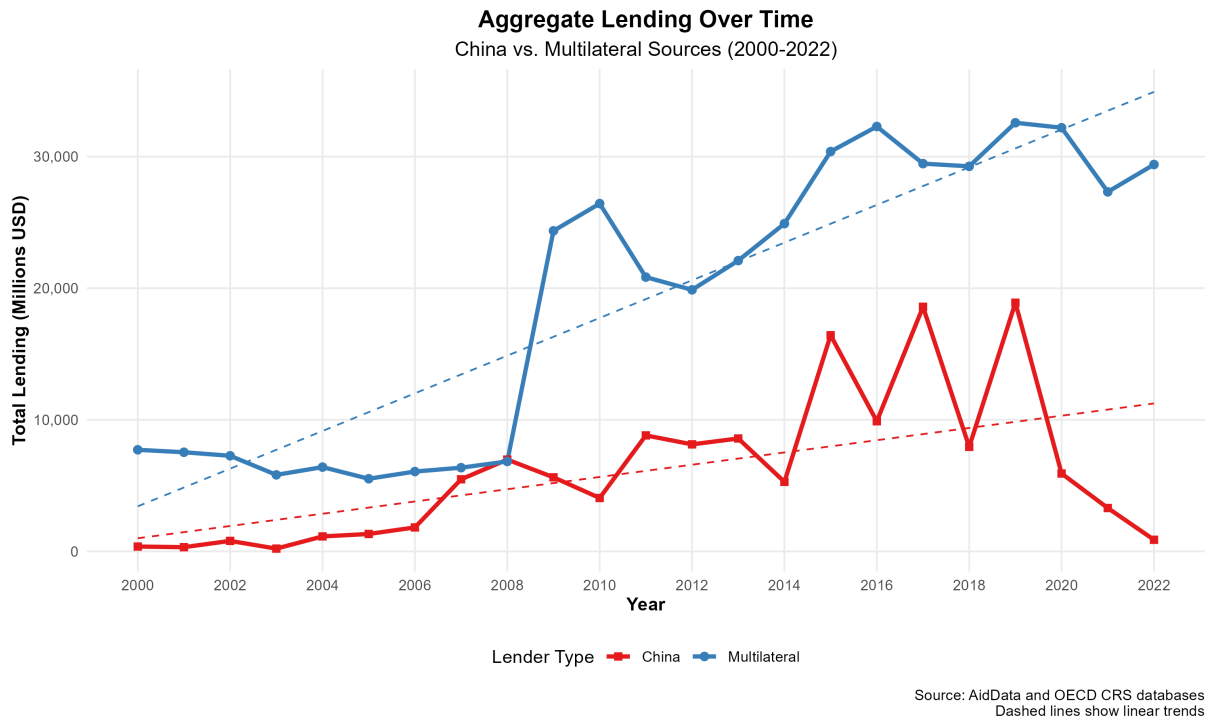


Table 5: Top 10 Recipients of Chinese and Multilateral Loans (2000-2022)

Chinese Loans			Multilateral Loans		
Country	Rank	Amount	Country	Rank	Amount
Sri Lanka	1	4,593	India	1	35,535
Angola	2	4,045	Turkey	2	33,047
Pakistan	3	3,950	Brazil	3	28,256
Vietnam	4	3,444	Indonesia	4	23,760
Bangladesh	5	3,350	Mexico	5	20,408
Indonesia	6	3,069	Egypt	6	19,042
Turkey	7	3,057	Argentina	7	17,402
Cambodia	8	3,050	Colombia	8	17,228
Congo, Rep.	9	3,027	Morocco	9	16,391
Ethiopia	10	2,891	Ukraine	10	15,176

Note: All amounts in millions of constant 2023 USD. Indonesia and Turkey appear in both lists.

Table 6: Loan Concentration: Herfindahl-Hirschman Index

Loan Source	HHI	Concentration Level
Chinese Loans	257.03	Unconcentrated
Multilateral Loans	443.04	Unconcentrated

Note: $HHI < 1,500$ = Unconcentrated; $1,500-2,500$ = Moderately concentrated; $> 2,500$ = Highly concentrated

Model Details and Robustness

Table 7: Mundlak Test for Fixed vs Random Effects (HC3 Robust Standard Errors)

Variable	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	11.2740	3.8253	2.947	0.0033**
<i>ChinaLoans</i>	-0.0007	0.0021	-0.328	0.743
<i>MultilateralLoans</i>	0.0010	0.0017	0.558	0.577
<i>Governance</i>	2.3060	0.9374	2.460	0.0140*
<i>ChinaLoans</i> $\times$ <i>Governance</i>	-0.0010	0.0005	-2.009	0.0447*
<i>MultilateralLoans</i> $\times$ <i>Governance</i>	-0.0013	0.0012	-1.039	0.299
<i>GDP_Growth</i>	-0.1528	0.0687	-2.222	0.0264*
<i>Inflation</i>	-0.0065	0.0792	-0.081	0.935
<i>Trade_GDP_ratio</i>	-0.0998	0.0210	-4.750	2.22e-06***
<i>Debt_GNI_Ratio</i>	0.1112	0.0212	5.249	1.74e-07***
<i>TotalReserves</i>	-0.0414	0.0125	-3.306	0.00097***
Between Effects:				
<i>Governance_mean</i>	-1.2525	1.0282	-1.218	0.223
<i>ChinaLoans_mean</i>	0.0082	0.0178	0.463	0.644
<i>MultilateralLoans_mean</i>	0.0109	0.0033	3.322	0.00092***
<i>GDP_Growth_mean</i>	-0.6687	0.6091	-1.098	0.272
<i>Inflation_mean</i>	0.0161	0.1733	0.093	0.926
<i>Trade_mean</i>	0.0106	0.0258	0.411	0.681
<i>Debt_GNI_mean</i>	0.1084	0.0460	2.356	0.0186*
<i>TotalReserves_mean</i>	0.0390	0.0183	2.131	0.0333*
Observations	N = 1540, n = 80			

Significance codes: *** 0.001, ** 0.01, * 0.05

Note: Mundlak test including within and between effects. Highly significant between effect with *MultilateralLoans* suggests selection effects. While the within-country effect of *ChinaLoans* $\times$ *Governance* is statistically significant in this specification, this has almost no real-world impact because of its near-zero coefficient and effect size.

Table 8: Between Effects Regression (HC3 Robust Standard Errors)

Variable	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	10.3516	4.3099	2.402	0.0189*
<i>ChinaLoans_mean</i>	-0.0067	0.0247	-0.271	0.787
<i>MultilateralLoans_mean</i>	0.0140	0.0070	2.003	0.0491*
<i>Governance_mean</i>	1.1235	0.7446	1.509	0.136
<i>ChinaLoans</i> × <i>Governance_mean</i>	-0.0057	0.0091	-0.626	0.534
<i>MultilateralLoans</i> × <i>Governance_mean</i>	0.0001	0.0059	0.022	0.982
<i>GDP_Growth_mean</i>	-0.6231	0.6515	-0.957	0.342
<i>Inflation_mean</i>	0.0332	0.2026	0.164	0.870
<i>Trade_GDP_mean</i>	-0.0680	0.0234	-2.909	0.0048**
<i>Debt_GNI_mean</i>	0.1949	0.0486	4.012	0.00015***
<i>TotalReserves_mean</i>	-0.0069	0.0108	-0.635	0.528
Observations	N = 80 (cross-sectional units)			

Significance codes: *** 0.001, ** 0.01, * 0.05

Note: Between effects (cross-sectional) regression using country means. Multilateral loans, trade/GDP ratio, and debt/GNI ratio show significant effects, indicating their importance in explaining cross-country variation in debt service ratios.

Table 9: Random Effects Regression Results (HC3 Robust Standard Errors)

Variable	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	20.2820	2.6876	7.546	7.65e-14***
<i>ChinaLoans</i>	-0.00002	0.0031	-0.006	0.995
<i>MultilateralLoans</i>	0.00113	0.0029	0.394	0.694
<i>Governance</i>	1.8322	0.7237	2.532	0.0114*
<i>ChinaLoans</i> × <i>Governance</i>	-0.00033	0.00056	-0.597	0.551
<i>MultilateralLoans</i> × <i>Governance</i>	-0.00148	0.00180	-0.824	0.410
<i>GDP_Growth</i>	-0.1297	0.0927	-1.399	0.162
<i>Inflation</i>	-0.0111	0.0851	-0.130	0.896
<i>Trade_GDP_ratio</i>	-0.0993	0.0182	-5.463	5.47e-08***
<i>Debt_GNI_Ratio</i>	0.1174	0.0285	4.117	4.04e-05***
<i>TotalReserves</i>	-0.0345	0.0100	-3.445	0.00059***
Observations	N = 1540, n = 80, T = 2-22 (Unbalanced Panel)			
R² / Adj. R²	0.2366 / 0.1430			
Total SS / Residual SS	99,116 / 75,668			

Significance codes: *** 0.001, ** 0.01, * 0.05

Note: Random effects specification estimated using HC3 standard errors. Results show consistent patterns with the main FE specification, with economic and institutional fundamentals remaining highly significant predictors of debt service ratios.

Table 10: Robustness Check: Governance Threshold Effects (HC3 Standard Errors)

Variable	Estimate	Std. Error	t value	Pr(> t)
<i>ChinaLoans</i>	-0.00007	0.00229	-0.032	0.975
<i>HighGovernance</i>	-0.2383	1.3866	-0.172	0.864
<i>LowGovernance</i>	-0.1009	2.2945	-0.044	0.965
<i>MultilateralLoans</i>	0.00187	0.00141	1.319	0.187
<i>ChinaLoans</i> × <i>HighGov</i>	0.00215	0.00325	0.662	0.508
<i>ChinaLoans</i> × <i>LowGov</i>	-0.00038	0.01128	-0.033	0.973
<i>MultilateralLoans</i> × <i>HighGov</i>	0.00175	0.00250	0.699	0.485
<i>MultilateralLoans</i> × <i>LowGov</i>	-0.00225	0.00439	-0.512	0.608
<i>GDP_Growth</i>	-0.1366	0.07023	-1.945	0.052
<i>Inflation</i>	-0.0273	0.08246	-0.331	0.741
<i>Trade_GDP_ratio</i>	-0.0991	0.02366	-4.189	2.98e-05***
<i>Debt_GNI_Ratio</i>	0.1145	0.02155	5.312	1.26e-07***
<i>TotalReserves</i>	-0.0364	0.01255	-2.905	0.0037**
Observations	N = 1540, n = 80, T = 2-22 (Unbalanced Panel)			
R² / Adj. R²	0.2368 / 0.1433			

Significance codes: *** 0.001, ** 0.01, * 0.05

Note: Threshold analysis using tertile splits (top/bottom third of governance scores). This governance approach leaves governance quality completely insignificant, implying that governance effects operate on a continuous spectrum.

Table 11: Principal Component Analysis of Worldwide Governance Indicators

Indicator	PC1	PC2	PC3	PC4	PC5	PC6
Government Effectiveness	0.423	0.273	-0.277	0.105	0.808	-0.074
Control of Corruption	0.422	0.103	-0.207	-0.704	-0.191	0.485
Rule of Law	0.432	0.055	-0.093	-0.177	-0.328	-0.814
Voice and Accountability	0.386	-0.063	0.907	-0.048	0.141	0.051
Regulatory Quality	0.415	0.388	-0.066	0.639	-0.427	0.287
Political Stability	0.367	-0.870	-0.212	0.227	0.009	0.107

Component	SD	Proportion of Variance	Cumulative Proportion
PC1	2.260	0.851	0.851
PC2	0.621	0.064	0.915
PC3	0.536	0.048	0.963
PC4	0.347	0.020	0.983
PC5	0.226	0.008	0.992
PC6	0.223	0.008	1.000

Note: The first principal component (PC1) explains 85.1% of the total variance in the six Worldwide Governance Indicators. All six indicators load positively and substantially on PC1 (loadings: 0.367–0.432). PC1 was retained as the composite governance index for all analyses.

Table 12: Correlation Matrix in Main Specification

	DSR	China	Multi.	Gov.	China×Gov.	Multi.×Gov.	GDP	Infl.	Trade	Debt/GNI	Reserves
Debt Service Ratio	1.000	0.002	0.311	0.132	0.010	-0.186	-0.165	0.083	-0.178	0.428	-0.294
China Loans	0.002	1.000	0.081	-0.118	-0.679	-0.154	0.078	0.020	-0.065	-0.073	-0.021
Multilateral Loans	0.311	0.081	1.000	0.128	-0.035	-0.672	-0.111	0.054	-0.270	-0.125	-0.043
Governance	0.132	-0.118	0.128	1.000	0.167	0.176	-0.077	-0.204	0.132	0.044	0.102
China×Governance	0.010	-0.679	-0.035	0.167	1.000	0.123	-0.060	-0.007	0.026	0.056	-0.019
Multi.×Governance	-0.186	-0.154	-0.672	0.176	0.123	1.000	0.076	-0.105	0.199	0.104	0.040
GDP Growth	-0.165	0.078	-0.111	-0.077	-0.060	0.076	1.000	-0.036	0.072	-0.141	-0.014
Inflation	0.083	0.020	0.054	-0.204	-0.007	-0.105	-0.036	1.000	-0.064	0.093	-0.117
Trade/GDP	-0.178	-0.065	-0.270	0.132	0.026	0.199	0.072	-0.064	1.000	0.318	0.021
Debt/GNI	0.428	-0.073	-0.125	0.044	0.056	0.104	-0.141	0.093	0.318	1.000	-0.376
Total Reserves	-0.294	-0.021	-0.043	0.102	-0.019	0.040	-0.014	-0.117	0.021	-0.376	1.000

Note: No severe multicollinearity issues, with the largest coefficients between loan-source variables and their own interaction terms. The highest non-interaction coefficient is between *Multilateral Loans* and *DSR*, which, as discussed in the paper, is likely a result of selection bias.

Table 13: Robustness Checks: Key Coefficients Across Specifications

Specification	Coefficient (Standard Error)				
	Governance	China	Multi.	China×Gov	Multi.×Gov
Baseline (1-year lag)	1.8921* (0.9263)	0.00003 (0.0020)	0.00066 (0.0015)	-0.00034 (0.00045)	-0.00152 (0.00112)
No lag	2.0259* (0.9232)	0.00081 (0.0017)	-0.00134 (0.0016)	0.00020 (0.00038)	-0.00168 (0.00114)
2-year lag	2.0884* (0.9976)	-0.00030 (0.0008)	-0.00048 (0.0012)	-0.00052 (0.00060)	-0.00273* (0.00130)
Loans/GDP	1.7075 (0.9708)	0.08200 (0.1491)	1.1368 (0.5990)	-0.00045 (0.00036)	-0.00162 (0.00087)
Post-BRI (2013-2022)	1.9059 (1.8864)	0.00049 (0.0031)	0.00112 (0.0020)	0.00017 (0.00053)	-0.00245 (0.00165)
Pre-BRI (2000-2012)	1.2466 (1.0493)	-0.00118 (0.0022)	-0.00308** (0.0011)	-0.00206 (0.00133)	-0.00002 (0.00172)

Sample sizes: Post-BRI: N=737; Pre-BRI: N=803; Others: N=1540
Note: All models include country/year FE with HC3 standard errors.
Control variables omitted from table for brevity, as they did not notably deviate between specifications.
Significance codes: *** 0.001, ** 0.01, * 0.05

Table 14: Regional Heterogeneity: Key Patterns

Region	Chinese Loans	China×Region	Significance
Europe & Central Asia	Negative coefficient	Positive interaction coeff.	$p = 0.034^*$
Middle East & North Africa	Negative coefficient	Positive interaction coeff.	$p = 0.015^*$
Sub-Saharan Africa	Negative coefficient	No significant interaction	$p = 0.092$
Latin America & Caribbean	Negative coefficient	No significant interaction	$p = 0.326$
South Asia	Negative coefficient	No significant interaction	$p = 0.451$

Multilateral loans: No significant regional interactions (all $p > 0.29$)

Sample: N=1540; All models use controls and fixed effects analogous to the baseline specification

Note: These results were derived from a variation of the main FE specification with added interaction terms between the loan-source variables and each region, respectively. The results listed in this table discuss the coefficients and significance of the Chinese loan term, as all multilateral terms were statistically insignificant. The main *ChinaLoans* and *MultilateralLoans* variables remained insignificant as well, with p-values of $p = 0.06$ and $p = 0.91$, respectively.

Eigenständigkeitserklärung

Ich versichere hiermit, dass ich die vorstehende Bachelorarbeit selbstständig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel benutzt habe, dass die vorgelegte Arbeit noch an keiner anderen Hochschule zur Prüfung vorgelegt wurde und dass sie weder ganz noch in Teilen bereits veröffentlicht wurde. Wörtliche Zitate und Stellen, die anderen Werken dem Sinn nach entnommen sind, habe ich in jedem einzelnen Fall kenntlich gemacht.

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Bonn, den 14.01.2026